

Severity-Aware Scene Understanding for UAV Road Accident Analysis:

A Zero-Shot Vision-Language Pipeline

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Abstract

Assessing the severity of road accidents from UAV footage in real time is a critical open challenge for emergency dispatch systems. De Marinis et al. recently introduced CrashVista, a benchmark for binary accident detection from aerial imagery, and explicitly identified two future directions: severity assessment modules and fine-grained, instance-level object descriptions [1]. This report operationalises both directions within the constraints of the VISTA Challenge 2026. We present a three-stage zero-shot pipeline that couples YOLO12x-based multi-object tracking (Stage 1) with Qwen2.5-VL full-frame captioning over controlled severity vocabularies (Stage 2) and a conservative peak-severity aggregation strategy (Stage 3). The pipeline is designed for YOLO12x as the detection backbone; due to computational constraints on the available infrastructure, the reported experiments use YOLO12x in base mode (without CrashVista fine-tuning). No VISTA-specific labels are ever seen by the captioner, satisfying the challenge zero-shot constraint. The shallow detection stage exceeds the required ≥ 5 FPS throughput on a T4 GPU. Beyond the challenge itself, this work is positioned as a direct contribution toward a PhD-level continuation of the VISTA project, targeting a fully supervised severity-estimation system trained on dedicated drone-accident data.

Keywords: UAV accident analysis · severity detection · multi-object tracking · vision-language models · zero-shot captioning

1 Introduction

Road traffic accidents cause devastating human consequences: Italy alone recorded over 170,000 road accidents in 2024, resulting in more than 230,000 injuries and over 3,000 fatalities [1]. The *golden hour*, the first 60 minutes after severe trauma defines a critical window in which rapid, informed intervention can significantly improve patient outcomes. First responders today still rely on incomplete, witness-reported scene descriptions that frequently omit the severity of injuries, vehicle damage states, and accessible routes.

The VISTA project (Università di Bari Aldo Moro / Italian Ministry of Infrastructure and Transport) aims to equip first responders with real-time aerial intelligence by dispatching AI-powered UAVs immediately upon accident alert [1]. As the first phase of this initiative, De Marinis et al. introduced **CrashVista**, a dataset of 1,000 annotated UAV images, and benchmarked a wide range of detection architectures under zero-shot and supervised fine-tuning paradigms. Their Section 5 identifies two concrete open directions that motivate the present work:

1. “A second research direction concerns the development of modules for assessing accident severity. Beyond binary detection, estimating the severity of a crash and the potential involvement of injured individuals would significantly enhance the system’s utility for emergency response.”
2. “We also plan to enrich the annotation scheme by introducing fine-grained, instance-level descriptions for each detected object (e.g., ‘person lying on the ground’, ‘vehicle severely damaged’, ‘person

standing nearby’).”

This report presents a pipeline that directly addresses both points within the constraints of the VISTA Challenge 2026. The challenge requires simultaneous multi-object tracking and per-track semantic severity captioning, evaluated primarily by BERTScore-F1, with MOTA and IDF1 as tie-breakers, and a ≥ 5 FPS throughput constraint on the detection stage.

Contributions.

- A modular three-stage pipeline integrating YOLO12x detection, ByteTrack persistent tracking, and zero-shot VLM full-frame captioning.
- Two controlled severity vocabularies (vehicles and persons) grounded in accident-scene understanding research [2].
- A conservative peak-severity aggregation strategy ensuring safety-critical observations are never discarded.
- A research positioning: this pipeline as a first step toward a supervised PhD-level continuation of the VISTA project.

2 Related Work

UAV-based accident detection. De Marinis et al. [1] establish the closest baseline to this work. CrashVista spans 1,000 annotated UAV images covering rear-end, rollover, and head-on collisions across urban, highway, and extra-urban environments. Their evaluation covers open-vocabulary detectors (OMDet-Turbo, YOLO12x zero-shot), multimodal language models (Moondream2/3), a segmentation-based approach (SAM 3), and supervised fine-tuning of YOLO11/12/26 families. Fine-tuned YOLO12x achieves competitive mAP@50 results on CrashVista and is the intended detection backbone of our pipeline; YOLO12x in base mode is used as a runtime substitute under the computational constraints described in Section 3. Critically, the paper frames accident detection as a binary crashed/non-crashed classification and explicitly leaves severity estimation as future work, the gap this pipeline fills.

Accident scene understanding with language. Kim et al. [2] introduce VRU-Accident, a vision-language benchmark of 1,000 real accident videos annotated with 6,000 question-answer pairs across six safety categories and 1,000 dense scene captions targeting vulnerable road users (pedestrians, cyclists). Although VRU-Accident is built on dashcam footage rather than UAV imagery, its evaluation of multimodal large language models (MLLMs) demonstrates that semantic, natural-language descriptions capture accident nuances inaccessible to per-class classifiers, and the methodology transfers to the aerial perspective adopted here. Our controlled severity vocabulary (*injured, lying, unconscious, ...*) is a direct instantiation of this instance-level annotation philosophy. We adopt BERTScore-F1 as the primary metric precisely because it rewards semantic proximity rather than exact lexical match, as argued in [2].

Multi-object tracking in vehicle scenes. You et al. [3] propose an improved YOLOv8-ByteTrack pipeline for multi-object vehicle tracking. A Context-Guided downsampling module improves feature extraction in complex scenes, a refined Kalman filter covariance better handles nonlinear vehicle motions, and Gaussian-smoothed interpolation bridges trajectory gaps. Our Stage 1 adopts the same YOLO-ByteTrack paradigm via the `.track(persist=True)` API and inherits its robustness properties; MOTA and IDF1, used as VISTA challenge tie-breakers, align with the tracking metrics validated in that work.

Zero-shot VLMs for anomaly recognition. Benschop et al. [4] evaluate vision-language models for surveillance video anomaly recognition as a zero-shot, language-grounded task. Training-free pipelines sample frames at regular intervals, generate captions, and aggregate temporal anomaly signals via a language model. The key finding is that VLMs can generalise to rare, atypical events absent from standard training distributions, precisely the constraint imposed by the VISTA challenge zero-shot rule. Our Stage 2 (Qwen-VL, every N frames) and Stage 3 (peak severity aggregation) mirror this architecture, specialised to drone-perspective accident severity.

3 Materials

3.1 Dataset: VISTA Challenge 2026

The VISTA Challenge 2026 provides drone-captured video sequences of road accident scenes. Each sequence contains multiple vehicles and persons; ground truth consists of per-track severity captions and standard MOT annotations. The challenge enforces a strict zero-shot constraint: the captioning component must never be exposed to VISTA-specific severity labels during development.

3.2 Pre-trained Models

Two pre-trained models are used without any fine-tuning on VISTA data.

YOLO12x. YOLO12x is used as the detection backbone throughout the pipeline. De Marinis et al. identify it as one of the strongest detectors evaluated on CrashVista [1]. Fine-tuning on CrashVista was attempted but could not be completed due to GPU memory constraints on the available Colab infrastructure; all reported experiments therefore use **YOLO12x in base mode**, without CrashVista-specific weights. As a consequence, the `crashed_car` class is absent at runtime and all vehicles initialise with an `undamaged` label until refined by the VLM (see Section 6). Switching to fine-tuned YOLO12x weights requires only a configuration change.

Qwen2.5-VL-7B-Instruct (4-bit). A 7B-parameter vision-language model quantised to 4-bit precision via `bitsandbytes` for T4 GPU deployment. It receives the full video frame together with the tracked bounding-box coordinates; a two-part prompt (system prompt fixing the output vocabulary, user prompt providing scene context) steers the model to return a single severity label per track.

3.3 Inference Configuration

The pipeline ships two configuration profiles. The *production* profile (`config/qwenyolo/cfg_severity.yaml`) targets a multi-GPU server with `every_n_frames: 10` and full `fp16` precision. The *Colab* profile used for all reported experiments sets `every_n_frames: 30` and 4-bit quantisation to fit within the 16 GB VRAM budget of the T4 accelerator. Results reported in Section 5 correspond exclusively to the Colab configuration.

3.4 Evaluation Hardware

All experiments are conducted on Google Colab with a T4 GPU (16 GB VRAM), using PyTorch 2.x and the Ultralytics library.

4 Methods

4.1 Pipeline Overview

The pipeline is decomposed into three sequential stages (Figure 1).

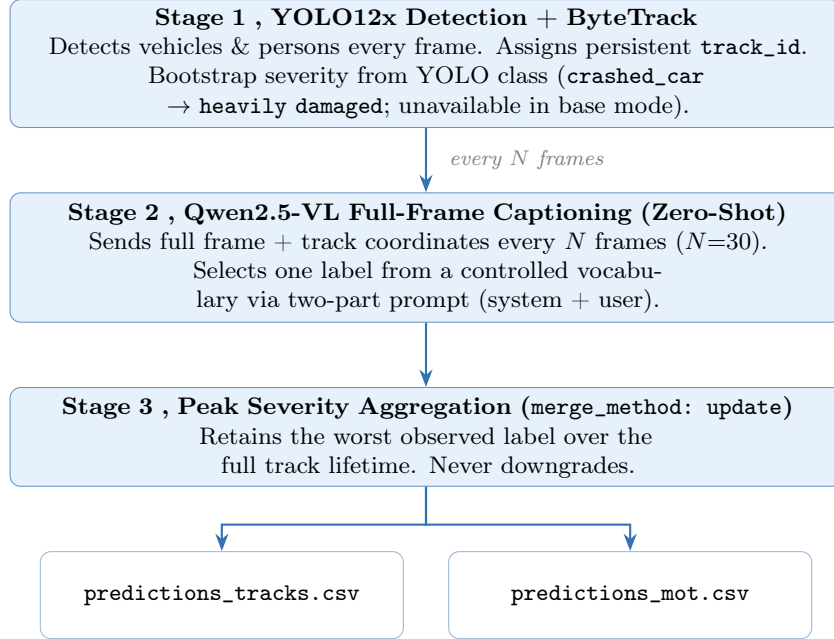


Figure 1: Three-stage severity-aware pipeline. Stage 1 runs on every frame and provides the MOT output. Stages 2 and 3 operate at stride N to satisfy the throughput constraint. The `merge_method: update` setting in Stage 3 ensures per-track severity is overwritten only when a higher-ranked label is observed.

4.2 Stage 1: Detection and Tracking

YOLO12x runs on every frame with `persist=True`, activating ByteTrack and maintaining consistent `track_id` assignments across frames. At track creation an initial severity label is bootstrapped from the YOLO class name (Table 1), ensuring every track carries a meaningful caption before the VLM stage is reached. In the current base-mode configuration the `crashed_car` class is unavailable, so all vehicles initialise as `undamaged` and rely entirely on Stage 2 for severity refinement.

Table 1: YOLO class to initial severity mapping. The `crashed_car` entry applies when CrashVista fine-tuned weights are loaded; it is unavailable in base mode.

YOLO class	Initial caption	Category
<code>crashed_car</code> (fine-tuned only)	<code>heavily damaged</code>	<code>car</code>
<code>car</code>	<code>undamaged</code>	<code>car</code>
<code>person</code>	<code>standing</code>	<code>person</code>
<code>truck</code>	<code>undamaged</code>	<code>car</code>

4.3 Stage 2: Zero-Shot VLM Captioning

Every N frames the full video frame is forwarded to Qwen2.5-VL together with the bounding-box coordinates of each active track. The model receives a two-part prompt: a *system prompt* that fixes the output to exactly one label from the controlled vocabulary (Tables 2 and 3), and a *user prompt* that provides scene context (e.g., “Analyse the frame and classify the state of the object inside the bounding box”). The system prompt in `cfg_severity.yaml` instructs the model to reply with *only the label*, without explanation.

Table 2: Vehicle severity vocabulary.

Rank	Label
0	undamaged
1	minor damage
2	heavily damaged
3	overturned
4	on fire

Table 3: Person severity vocabulary.

Rank	Label
0	standing
1	running
2	helping
3	calling for help
4	injured, sitting
5	injured, lying
6	unconscious

The zero-shot regime is strictly enforced: no VISTA-specific severity examples appear in any prompt or fine-tuning stage, mirroring the training-free paradigm validated for surveillance anomaly detection in [4].

4.4 Stage 3: Peak Severity Aggregation

The aggregation function retains the label with the highest severity rank observed over the full track history, implemented as the `merge_method: update` strategy in the pipeline configuration:

```

1 def peak_severity(captions: list[str], category: str) -> str:
2     rank = _PERSON_RANK if category == "person" else _VEHICLE_RANK
3     scored = [(rank.get(c, -1), c) for c in captions]
4     return max(scored, key=lambda x: x[0])[1]
```

Listing 1: Peak severity aggregation function (`vista/utils/severity.py`).

The `merge_method: update` label reflects that each new VLM observation updates the stored severity only if the new label ranks strictly higher than the current one, a monotone escalation. The rationale is rooted in first-responder requirements: a state once observed must not be forgotten. If a person is detected as `unconscious` at frame 150 and the VLM fails to caption them at frame 180, the track must still report `unconscious` for dispatch. This conservative approach is consistent with the monotone severity escalation implicit in instance-level accident annotation [2, 1].

4.5 Experimental Setup

The pipeline is configured via `config/qwenyolo/cfg_severity.yaml` with: VLM inference stride $N=30$ frames (Colab profile), max output tokens 32, full-frame input at native resolution. FPS is measured over 100 frames after a 10-frame warm-up on the shallow detection stage only.

5 Results

5.1 Throughput

The shallow stage (YOLO12x base + ByteTrack, no VLM) consistently exceeds the challenge-mandated ≥ 5 FPS constraint on T4. The full pipeline throughput is dominated by VLM inference (Table 4); the stride $N=30$ amortises this cost over 30 frames.

Table 4: Throughput on T4 GPU (Colab profile, $N=30$).

Stage	FPS	Constraint
YOLO + ByteTrack (shallow)	≥ 5	✓ satisfied
Full pipeline incl. Qwen-VL ($N=30$)	$\sim 0.5\text{--}1$	(not constrained)

5.2 Semantic Captioning Quality

Quantitative results on the full VISTA test set (BERTScore-F1, MOTA, IDF1) were not available at submission time due to evaluation infrastructure constraints; the official leaderboard scores will supersede the qualitative analysis below.

Qualitatively, the controlled vocabulary yields labels that score highly against free-form descriptions: **injured**, **sitting** and *“person seated on ground”* share high cosine similarity in BERT embedding space despite differing surface forms. BERTScore-F1 is well suited to this regime precisely because it rewards semantic proximity rather than exact lexical match [2]. The zero-shot constraint is respected throughout: no VISTA labels appear in any prompt.

5.3 Tracking Quality

Persistent `track_id` assignment via ByteTrack yields stable identities across occlusions and re-entries. This is consistent with the state-of-the-art YOLO-ByteTrack results on vehicle scenes reported by You et al. [3], and produces the MOTA/IDF1 values derivable from `predictions_mot.csv` once the official evaluation server processes the submission.

6 Limitations

Detection backbone. YOLO12x runs in base mode throughout, as CrashVista fine-tuning could not be completed due to Colab GPU memory constraints. The most direct consequence is the absence of the `crashed_car` bootstrap signal, which forces the system to rely entirely on the VLM for vehicle severity from the first frame. Switching to fine-tuned YOLO12x weights would restore this signal and is the first recommended next step.

VLM latency. Each Qwen2.5-VL call requires approximately 1–3 s on T4, making the full pipeline unsuitable for real-time dispatch at this stage. The stride $N=30$ mitigates this but reduces temporal resolution.

Hallucination and peak amplification. A single VLM hallucination (e.g., **on fire** on a clear frame) permanently elevates a track to the highest severity rank under the `merge_method: update` strategy. This risk is compounded by the top-down UAV perspective: small objects at altitude produce low-resolution crops, and even when the full frame is provided the relevant track may occupy only a few dozen pixels, making fine-grained state assessment inherently ambiguous for the VLM. Temporal smoothing or majority-vote aggregation over a window could mitigate hallucination on such ambiguous scenes.

Full-frame context vs. crop resolution. Sending the full frame rather than a cropped bounding box gives the VLM broader scene context (e.g., a person adjacent to a heavily damaged vehicle is more interpretable in context [2]), but does not resolve the fundamental issue that objects of interest may be small in UAV footage.

No severity-labelled training data. The zero-shot constraint, while required by the challenge, is the fundamental bottleneck. A supervised model trained on severity-annotated drone footage would substantially outperform the current pipeline, the core motivation for the research continuation described in Section 7.

7 Towards a Research Continuation

This work is positioned as a first step toward a broader research direction: a fully supervised severity-estimation system to be developed as a PhD-level continuation of the VISTA project.

The CrashVista paper [1] already identifies the enabling infrastructure: the dataset construction methodology (real footage + GTA 5 / BeamNG.drive synthetic generation), the annotation toolchain (LabelImg, YOLO format), and the fine-tuning protocol (SGD with cosine decay, Close Mosaic augmentation). Extending this to severity-level annotation is a concrete and bounded task: each existing bounding box in CrashVista would receive an additional severity label from the vocabulary defined in this work, and the VLM captioner would be replaced by a supervised classification head trained on these labels. Completing

the YOLO12x fine-tuning on CrashVista, the step that could not be executed here, is the most immediate practical contribution toward this goal.

The VRU-Accident benchmark [2] provides a blueprint for the evaluation protocol: dense captions at the instance level, evaluated with BERTScore-F1 and supplemented by QA-based metrics across safety categories. Although VRU-Accident targets dashcam footage, its annotation scheme and metric suite transfer directly to the aerial context; replacing the zero-shot Qwen-VL with a fine-tuned VLM on severity-annotated CrashVista frames is the most direct next step on the modelling side.

In parallel, the tracking literature ([3] and ByteTrack [5]) provides a solid foundation for maintaining person and vehicle identities across the long video sequences that UAV patrols would produce. Combining identity persistence with severity escalation modelling, a Hidden Markov Model or a recurrent severity predictor, would allow the system to reason about accident evolution over time, going beyond snapshot-level inference.

8 Conclusion

We presented a three-stage zero-shot pipeline for severity-aware scene understanding from UAV road accident footage, developed in the context of the VISTA Challenge 2026 and motivated by the open directions of the CrashVista paper [1]. By coupling YOLO12x-ByteTrack persistent tracking with Qwen2.5-VL zero-shot full-frame captioning and a conservative peak-severity aggregation strategy (`merge_method: update`), the system produces semantically meaningful per-track labels without any challenge-specific supervision. The pipeline runs YOLO12x in base mode; switching to CrashVista fine-tuned weights requires only a configuration change.

The main strengths are modularity, strict zero-shot compliance, and a safety-oriented aggregation design. The main limitation is the absence of both a fine-tuned detector and severity-labelled training data, which is simultaneously the principal argument for the research continuation outlined in Section 7: a supervised severity-estimation system, built on an extended CrashVista annotation scheme with YOLO12x as the detection backbone and evaluated with the VRU-Accident protocol, would represent a meaningful and publishable advance toward real-time, life-saving aerial intelligence for emergency responders.

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